§ 39. Algorithm for the calculation of the roots.

The proof given in the preceding paragraph for the existence of a root of an algebraic equation, although it leaves nothing to be desired in conclusiveness, nevertheless still has the defect that it does not make clear the steps by which a root can be calculated by a convergent computational procedure. We therefore add the following considerations, intended to remedy this defect, even if only theoretically. They also give a second proof of the fundamental theorem, due in its essentials to Lipschitz (*Lehrbuch der Analysis*, vol. I, § 61 ff.; I owe a simplification to a communication of Dedekind).

If the two integral rational functions $f(x)$ and $f'(x)$ have a common divisor, then this can be found and removed by rational operations according to § 6. We may therefore assume that $f(x)$ and $f'(x)$ have no common divisor, that is, that they do not vanish simultaneously for any value of x.

We now assume that the calculation of the roots has already succeeded for a function of degree $n-1$, and accordingly suppose $f'(x)$ resolved into linear factors. Thus, if

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots, \tag{1}$$

then

$$f'(x) = nx^{n-1} + (n-1)a_1x^{n-2} + \cdots = n(x-\beta_1)(x-\beta_2)\cdots(x-\beta_{n-1}), \tag{2}$$

where $\beta_1, \beta_2, \ldots, \beta_{n-1}$ are to be regarded as known numbers, some of which may also be identical. By our assumption, the absolute values

$$|f(\beta_1)|, |f(\beta_2)|, \ldots, |f(\beta_{n-1})|, \tag{3}$$

which we denote by

$$b_1, b_2, \ldots, b_{n-1}, \tag{4}$$

are all different from zero; let the notation be so chosen that b_1 is the smallest among them, or at least exceeds none of the others. By theorem 5 of § 38 a value α of x can then be determined so that the absolute value a of $f(\alpha)$ is smaller than b_1, hence

$$a < b_1, b_2, \ldots, b_{n-1}. \tag{5}$$

We now bound a domain G for the variable x in such a way that outside this domain the absolute value of $f(x)$ is always greater than a, so that G contains all points x for which $|f(x)| < a$ (but also other points). By § 31, IV this is possible in the following way. About the origin as centre draw a circle (R) of sufficiently large radius R containing the points $\alpha, \beta_1, \beta_2, \ldots, \beta_{n-1}$, and cut out of this circle the points $\beta_1, \beta_2, \ldots, \beta_{n-1}$, at which $|f(x)|$ is certainly greater than a, by circular envelopes of sufficiently small but non-zero radii $\rho_1, \rho_2, \ldots, \rho_{n-1}$, so that within all these circles $(\rho_1), (\rho_2), \ldots, (\rho_{n-1})$ the absolute value $|f(x)|$ remains greater than a (§ 31, V). None of these circles then encloses the point α, for at that point $|f(x)| = a$. The connected piece of surface bounded by the circles $(R), (\rho_1), \ldots, (\rho_{n-1})$ is the domain G.

We now determine a sequence of numbers

$$\alpha, \alpha', \alpha'', \alpha''', \ldots$$

in such a way that the absolute values of $f(\alpha), f(\alpha'), f(\alpha''), \ldots$, which we denote by

$$a, a', a'', a''', \ldots,$$

always decrease. The points so determined all remain in the interior of the domain G, since for them $|f(x)| < a$.

For this we use a procedure quite similar to that used for the proof of theorem 5 in § 38, but simplified by the fact that $f'(\alpha)$ is already different from zero. If δ denotes a positive proper fraction, still to be determined more precisely, we put in the expansion

$$f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1 \cdot 2}f''(\alpha) + \cdots \tag{6}$$

$$h = -\delta\frac{f(\alpha)}{f'(\alpha)}, \tag{7}$$

and obtain

$$f(\alpha+h) = (1-\delta)f(\alpha) + \frac{\delta^2 f(\alpha)^2 f''(\alpha)}{1\cdot 2\, f'(\alpha)^2} - \frac{\delta^3 f(\alpha)^3 f'''(\alpha)}{1\cdot 2\cdot 3\, f'(\alpha)^3} + \cdots. \tag{8}$$

Now by (2)

$$f'(\alpha) = n(\alpha-\beta_1)(\alpha-\beta_2)\cdots(\alpha-\beta_{n-1});$$

therefore, if the absolute values of $\alpha-\beta_1, \alpha-\beta_2, \ldots, \alpha-\beta_{n-1}$ are denoted by $r_1, r_2, \ldots, r_{n-1}$,

$$|f'(\alpha)| = nr_1 r_2\cdots r_{n-1}.$$

Since α lies outside the circle of radius ρ_i drawn about β_i, we have $r_1 > \rho_1, r_2 > \rho_2, \ldots$, and consequently

$$|f'(\alpha)| > n\rho_1\rho_2\cdots\rho_{n-1}.$$

Thus $|f'(\alpha)|$ is greater than a positive number k independent of the position of α in G. It follows that one can choose a sufficiently large positive number Q, likewise independent of the position of α and of the proper fraction δ, so that

$$\left|\frac{f(\alpha)^2 f''(\alpha)}{1\cdot 2\, f'(\alpha)^2} - \delta\frac{f(\alpha)^3 f'''(\alpha)}{1\cdot 2\cdot 3\, f'(\alpha)^3} + \cdots\right| < Q. \tag{9}$$

One obtains Q, for example, by replacing in the denominators on the left of (9) $f'(\alpha)$ by k, replacing δ by 1, replacing all the other factors by their absolute values, replacing finally the absolute value of α by the greater value R, and then making the number so obtained arbitrarily larger. From (8) one obtains

$$|f(\alpha+h)| < (1-\delta)|f(\alpha)| + \delta^2 Q. \tag{10}$$

Assume now Q so chosen that, for every α in G,

$$2Q > |f(\alpha)| = a,$$

which is plainly compatible with the above determination. We may then put

$$\delta = \frac{a}{2Q}, \qquad h = -\frac{af(\alpha)}{2Qf'(\alpha)}, \tag{11}$$

and, denoting $\alpha+h$ by α', and denoting $|f(\alpha')|$ by a', obtain from (10)

$$a' < a\left(1-\frac{a}{4Q}\right). \tag{12}$$

By the same procedure we derive from α', a' a second system α'', a'', then from α'', a'' a third system α''', a''', and so on; for the sequence of decreasing positive numbers $a, a', a'', \ldots$ we obtain the inequalities

$$a' < a\left(1-\frac{a}{4Q}\right), \quad a'' < a'\left(1-\frac{a'}{4Q}\right), \quad a''' < a''\left(1-\frac{a''}{4Q}\right), \ldots. \tag{13}$$

We prove that this sequence approaches the limit zero. If this were not the case, a positive number ω could be given such that all $a^{(\nu)}$ remain above ω; the differences

$$\frac{a^{(\nu)}}{4Q}$$

would then all be greater than the positive proper fraction $\omega/(4Q)$, and from (13) it would follow that

$$a' < a\theta, \quad a'' < a'\theta, \quad a''' < a''\theta, \quad \dots,$$

where $0 < \theta < 1$. By multiplication one gets

$$a^{(\nu)} < a\theta^{\nu}. \tag{14}$$

This is a contradiction, since θ^{ν} becomes infinitely small as ν increases without bound. The result so obtained is expressed by

$$\lim a^{(\nu)} = 0. \tag{15}$$

If the inequalities (13) are written in the form

$$a - a' > \frac{a^2}{4Q}, \qquad a' - a'' > \frac{a'^2}{4Q}, \qquad \dots,$$

and added from the $(\mu+1)$-st to the $(\nu+1)$-st, one obtains

$$(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \dots + (a^{(\nu)})^2 < 4Q\big(a^{(\mu)} - a^{(\nu+1)}\big),$$

and therefore, when μ and ν both tend to infinity,

$$\lim\big((a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \dots + (a^{(\nu)})^2\big) = 0. \tag{16}$$

For the sequence of numbers $\alpha, \alpha', \alpha'', \dots$ the law of formation resulting from (11) is

$$\begin{aligned} \alpha' &= \alpha - \frac{af(\alpha)}{2Qf'(\alpha)}, \\ \alpha'' &= \alpha' - \frac{a'f(\alpha')}{2Qf'(\alpha')}, \\ &\vdots \\ \alpha^{(\nu)} &= \alpha^{(\nu-1)} - \frac{a^{(\nu-1)}f(\alpha^{(\nu-1)})}{2Qf'(\alpha^{(\nu-1)})}. \end{aligned} \tag{17}$$

If an arbitrary number of these equations are added, from the $(\mu+1)$-st to the $(\nu+1)$-st, then

$$\alpha^{(\mu)} - \alpha^{(\nu+1)} = \sum_{r=\mu}^{\nu} \frac{a^{(r)}f(\alpha^{(r)})}{2Qf'(\alpha^{(r)})}. \tag{18}$$

Since all the $|f'(\alpha^{(r)})|$ are at least equal to k, it follows for the absolute value of this difference that

$$|\alpha^{(\mu)} - \alpha^{(\nu+1)}| < \frac{(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \dots + (a^{(\nu)})^2}{2Qk}, \tag{19}$$

and hence, by (16),

$$\lim |\alpha^{(\mu)} - \alpha^{(\nu+1)}| = 0. \tag{20}$$

This says (according to the concept of a sequence of numbers explained in the Introduction, p. 15) that the numbers $\alpha, \alpha', \alpha'', \dots$ approach a definite limit, which we shall denote by ξ. From the continuity of the function $f(x)$ it then follows easily that $f(\xi) = 0$; for if $|f(\xi)| > 0$, then, since the $\alpha^{(\nu)}$ come arbitrarily close to ξ, the absolute values $a^{(\nu)}$ of $f(\alpha^{(\nu)})$ could not sink below every positive value, which is precisely what we have proved for them.